

MEROMAIDAN

Nepal's Smart Sports Venue Booking and Management Platform

CONSOLIDATED PRODUCT REQUIREMENTS DOCUMENT (PRD)

Complete but concise implementation-ready edition

Document version	2.0 - Consolidated Draft
Document status	Ready for stakeholder review
Product type	Sports marketplace and multi-tenant SaaS
Primary stack	PHP 8.x, Laravel 12, MySQL, REST API
Prepared by	MeroMaidan Development Team
Classification	Confidential - Internal Project Document

Purpose

This document replaces the repetitive long draft with one structured PRD covering the complete product scope, role permissions, functional requirements, detailed user scenarios, data and API requirements, security, testing, and acceptance criteria.

Document Control

Item	Details
Source basis	Uploaded MeroMaidan PRD draft and Project Brief.
Consolidation approach	Duplicate sections removed; numbering corrected; requirements grouped by module.
Additional specification	Concise implementation-level data, API, UI/UX, testing, deployment, and field-admin requirements added to complete the PRD.
Review requirement	Business policies, payment provider, subscription packages, refund rules, and legal retention periods require stakeholder confirmation.

Revision History

Version	Status	Description
1.0	Original draft	Large chapter-based PRD prepared in multiple parts.
2.0	Consolidated draft	Shorter, complete PRD with detailed scenarios and implementation requirements.

Approval Roles

Role	Approval responsibility
Business sponsor	Confirms scope, business model, pricing, and operational policies.
Product owner	Confirms priorities, release scope, and acceptance criteria.
Technical lead	Confirms architecture, security, data, integration, and deployment feasibility.
Quality reviewer	Confirms traceability, testability, consistency, and document completeness.

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1. Executive Summary

MeroMaidan is a cloud-based sports venue marketplace and multi-tenant Software-as-a-Service platform designed for Nepal. It replaces fragmented booking through phone calls, social media messages, spreadsheets, and paper records with a centralized system for venue discovery, real-time availability, online booking, payment, customer engagement, and venue-business management.

The platform serves two connected markets. Players use the public marketplace to discover and reserve sports facilities. Sports businesses subscribe to a secure tenant workspace to manage venues, grounds, schedules, pricing, staff, customers, operational resources, content, subscriptions, reports, and analytics. A Super Admin portal governs the entire ecosystem.

Core product principle: Every tenant business operates in its own logically isolated workspace. All tenant-owned records must be protected by tenant-aware authorization and resource-ownership validation.

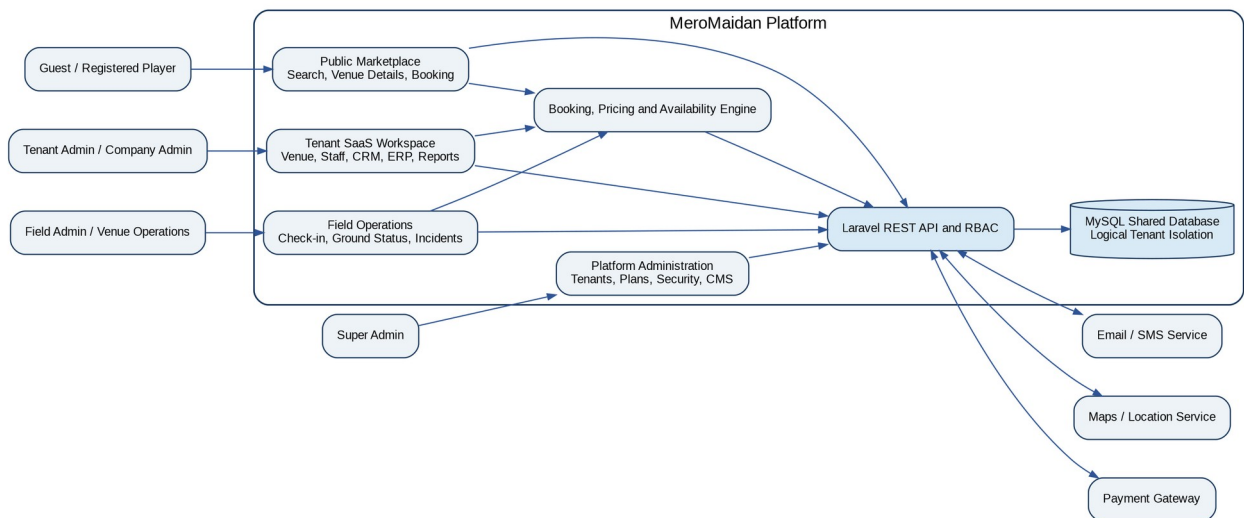


Figure 1. MeroMaidan platform ecosystem and major user groups

1.1 Product Summary

Area	Summary
Public marketplace	Venue discovery, filtering, details, real-time availability, booking, payment, booking history, favorites, reviews, and notifications.
Tenant SaaS	Business profile, venues, grounds, schedules, pricing, staff, booking operations, CRM, ERP, CMS, promotions, reports, analytics, subscription, and settings.
Field operations	Assigned venue dashboard, booking verification, customer check-in, ground readiness, operational status, no-shows, maintenance blocks, incidents, and daily summaries.
Super administration	Tenant verification, user and tenant governance, plans, payments, featured listings, CMS, analytics, security monitoring, audit logs, and support.

2. Product Vision, Mission and Goals

2.1 Vision

To become Nepal's most trusted and comprehensive digital platform for discovering, booking, and managing sports venues through a secure, scalable, and affordable multi-tenant SaaS ecosystem.

2.2 Mission

To simplify sports participation and venue operations by providing transparent availability, reliable online booking, secure payment, and practical business-management tools for players, venue owners, operational staff, and platform administrators.

2.3 Product Goals

ID	Goal	Definition
G-001	Improve venue accessibility	Enable users to discover venues by sport, location, facilities, pricing, rating, and availability.
G-002	Simplify booking	Provide a short, secure reservation journey while preventing conflicts and double bookings.
G-003	Digitize operations	Replace manual venue, ground, staff, booking, customer, and resource processes.
G-004	Deliver multi-tenant SaaS	Provide isolated workspaces while maintaining centralized platform governance.
G-005	Improve customer experience	Provide real-time information, confirmations, reminders, booking history, and policy-based cancellation.
G-006	Support business growth	Increase visibility, occupancy, customer retention, and data-driven decisions.
G-007	Create recurring revenue	Support configurable subscription plans and additional promotional revenue.
G-008	Enable long-term expansion	Use modular architecture that can support new sports, cities, applications, and integrations.

3. Problem Statement and Proposed Solution

3.1 Current Problems

- Double bookings and scheduling conflicts caused by manual calendars.
- No reliable real-time availability for customers.
- Venue information scattered across social media pages and private messages.
- Slow booking confirmation and inconsistent customer communication.
- Manual records with limited customer history and weak traceability.
- Limited online visibility for small and medium sports businesses.
- No integrated revenue, occupancy, cancellation, or customer-retention reporting.
- Difficulty managing multiple venues, grounds, staff members, prices, and maintenance schedules.

3.2 Proposed Solution

MeroMaidan provides one integrated ecosystem consisting of a public marketplace, a tenant SaaS workspace, a field-operations interface, and a Super Admin portal. Shared services such as authentication, authorization, booking, pricing, payment, notifications, audit logging, reporting, and analytics support all portals through secure REST APIs.

4. Product Scope and Release Strategy

4.1 Phase 1 Sports Categories

- Football grounds
- Futsal
- Cricket grounds
- Cricisal

4.2 Phase 1 In-Scope Capabilities

Domain	In-scope capability
Marketplace	Public pages, venue discovery, advanced filters, map/location search, venue details, ratings, and favorites.
Identity	Registration, email verification, login, password reset, role-based dashboards, session management, and account status.
Booking	Real-time availability, price calculation, booking validation, confirmation, modification, cancellation, refund flow, check-in, completion, and no-show.
Payments	Booking payments, subscription payments, invoices, receipts, refunds, transaction history, and reconciliation.
Tenant management	Vendor onboarding, approval, workspace provisioning, tenant settings, venues, grounds, pricing, availability, staff, CRM, ERP, CMS, reports, analytics, and subscriptions.
Field operations	Assigned venue schedule, check-in, booking verification, ground readiness, operational status, maintenance blocks, incidents, and daily summary.
Platform administration	Users, tenants, subscriptions, payments, advertisements, featured listings, CMS, analytics, security, configuration, support, and audit.
Technical services	REST API, tenant isolation, RBAC, notifications, logs, monitoring, backup, and secure file storage.

4.3 Out of Scope for Initial Release

- Native mobile applications
- AI-powered recommendations
- Tournament and league management
- Team and player community features
- Equipment rental marketplace
- Loyalty and reward programmes
- Corporate booking packages
- Demand-based dynamic pricing
- Multi-country deployment
- Advanced machine-learning analytics

4.4 Release Priorities

Priority	Meaning	Examples
Must	Required for launch	Authentication, tenant isolation, venue data, availability, booking, payment, tenant operations, field operations, Super Admin, audit logging.
Should	Important but can follow the core launch	Favorites, advanced reports, promotional campaigns, review moderation, cash-payment option, support tickets.
Could	Added when capacity permits	Referral offers, advanced exports, richer CMS, configurable custom roles.
Future	Explicitly outside Phase 1	Mobile apps, tournaments, AI, loyalty, community, equipment rental.

5. Stakeholders, Actors and Permissions

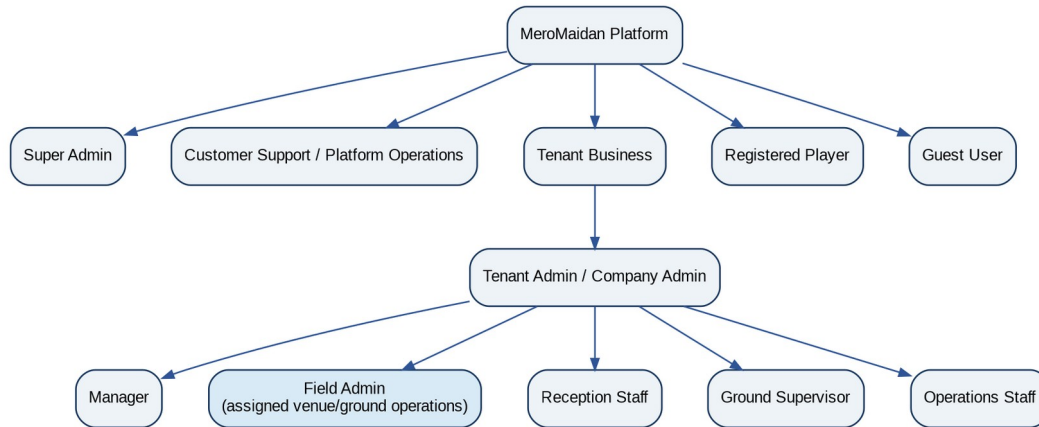


Figure 2. User and operational role hierarchy

5.1 Actor Definitions

Actor	Type	Primary responsibility
Guest User	Unauthenticated visitor	Browse, search, view public details, pricing, reviews, and register. Cannot book.
Registered Player	Verified customer	Book, pay, manage reservations, receive notifications, save favorites, and submit reviews.
Tenant Admin / Company Admin	Owner or authorized tenant administrator	Full control of tenant-owned business data, staff, venues, grounds, bookings, pricing, reports, and subscription.
Manager	Tenant operational manager	Supervise configured areas according to delegated permissions.
Field Admin	User-requested operational role aligned with manager/ground-supervisor duties	Manage day-to-day operations for assigned venues or grounds. No cross-tenant, platform, or unrestricted subscription access.
Reception Staff	Front-desk tenant staff	Verify bookings, check customers in, view schedules, and update permitted booking statuses.
Ground Supervisor	Ground-level operational staff	Confirm readiness, update ground status, and record maintenance or incidents.
Operations Staff	Tenant operational staff	Perform assigned venue and equipment tasks.
Super Admin	Platform administrator	Govern users, tenants, subscriptions, payments, content, security, support, and platform configuration.
External Service	Payment, messaging, and map provider	Provide payment processing, notifications, geolocation, and routing through controlled integrations.

5.2 High-Level Permission Matrix

Permission	Guest	Player	Tenant Admin	Manager	Field Admin	Super Admin
Browse marketplace	Yes	Yes	Yes	Assigned tenant	Assigned tenant	Platform-wide

Permission	Guest	Player	Tenant Admin	Manager	Field Admin	Super Admin
Create customer booking	No	Yes	Optional	Operational only	Operational only	Support exception
Manage business profile	No	No	Yes	Delegated	No	Platform oversight
Manage venues and grounds	No	No	Yes	Delegated	Assigned only	Oversight
Configure pricing	No	No	Yes	Delegated	No	Oversight
Manage staff and roles	No	No	Yes	Limited if delegated	No	Platform roles only
Verify and check-in booking	No	No	Yes	Yes	Yes	Exceptional support
Block ground / maintenance slot	No	No	Yes	Yes	Yes	Exceptional support
View tenant financial reports	No	Own payments	Yes	Limited if delegated	No	Platform-wide
Manage tenant subscription	No	No	Yes	No	No	Yes
View audit logs	No	No	Tenant scope	Limited	Operational entries	Platform-wide
Approve or suspend tenants	No	No	No	No	No	Yes

Field Admin boundary: Field Admin is a tenant-scoped operational role. It may be assigned to selected venues or grounds. It can manage operational status and booking execution but cannot change tenant ownership, subscription, platform configuration, or unrestricted financial settings.

6. Business Model and Success Metrics

6.1 Business Model

MeroMaidan follows a B2B2C model. The B2C marketplace serves players. The B2B SaaS product serves sports businesses. Platform revenue may come from subscriptions, featured listings, advertising, premium business services, and an optional booking commission if stakeholders approve it.

6.2 Subscription Approach

The source documents contain two example subscription structures: Free/Premium and Starter/Professional/Business/Enterprise. The production plan names and prices are not finalized. The application must therefore support configurable plans rather than hard-coded packages.

Configurable plan attribute	Examples
Limits	Venues, grounds, staff accounts, bookings, storage, API usage.
Features	Advanced analytics, CRM, ERP, marketing, exports, support level, featured visibility.
Billing	Monthly or annual price, trial, currency, tax, upgrade, downgrade, renewal, cancellation.
Enforcement	Feature access, limit checks, renewal reminders, expiry restriction, and reactivation.

6.3 Key Success Metrics

Metric	Definition
Active registered players	Monthly active verified customers.
Active tenants	Approved businesses with active workspaces.
Booking success rate	Successful confirmed bookings divided by initiated bookings.
Booking completion rate	Completed bookings divided by confirmed bookings.
Payment success rate	Successful payments divided by payment attempts.
Venue occupancy rate	Booked operational time divided by available operational time.
Cancellation and no-show rate	Cancelled/no-show bookings as a percentage of confirmed bookings.
Customer satisfaction	Average rating, review sentiment, and support resolution.
Vendor retention	Tenants renewing subscriptions.
Monthly recurring revenue	Recurring tenant subscription income.
Platform availability	Service uptime and error rate.
Support resolution time	Average duration to close support cases.

7. Product Architecture and Technology

7.1 Technology Baseline

Layer	Requirement
Backend	PHP 8.x with Laravel 12.
Frontend	Responsive web application using Next.js or React.js; final selection requires technical approval.
Database	MySQL with tenant-aware schema and logical isolation.
API	Versioned RESTful APIs using JSON over HTTPS.
Authentication	Laravel-supported secure authentication and verified user accounts.
Authorization	RBAC, policies, permissions, tenant context, and resource ownership checks.
Architecture	Modular multi-tenant SaaS with shared application and shared database.
Background processing	Queues and scheduled jobs for notifications, reconciliation, reports, reminders, and housekeeping.
Storage	Secure file and image storage with validation and access controls.
Monitoring	Centralized logs, metrics, alerting, security events, and audit records.

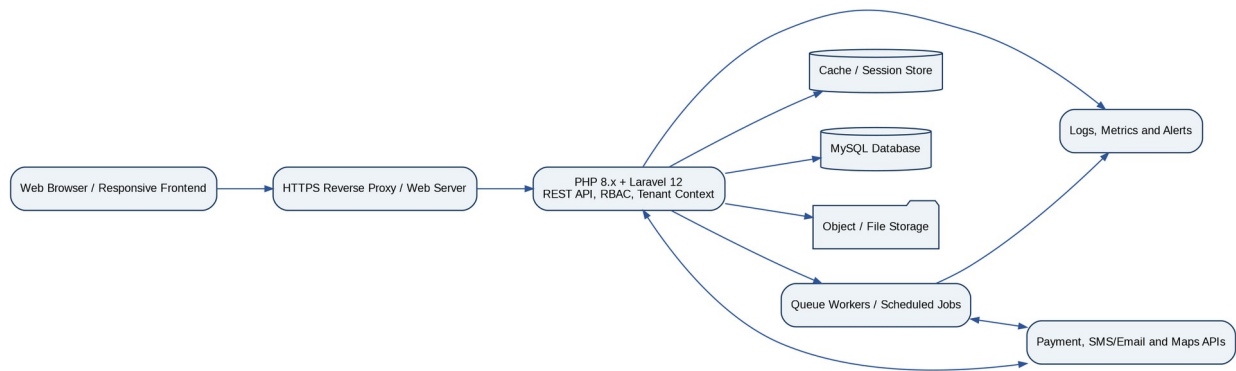


Figure 3. Recommended logical deployment architecture

7.2 Architectural Principles

- Tenant isolation: every tenant-owned query and command must use the active tenant context.
- Security by design: authentication, authorization, validation, encryption, audit, and monitoring are built into each module.
- Modularity: booking, payments, venues, users, subscriptions, notifications, reporting, and content remain logically separated.
- API-first integration: frontend and external services communicate through controlled, versioned endpoints.
- Traceability: critical status changes and administrative actions produce immutable audit events.
- Scalability: stateless application services and background workers should support horizontal scaling.
- Configuration over hard-coding: plans, policies, limits, sports, facilities, statuses, and notification templates should be configurable where practical.

8. Functional Modules and Requirements

All requirements use MoSCoW priority: Must, Should, Could, or Future. The word "shall" indicates a mandatory requirement for the stated release priority.

8.1 Identity and Access Management

Requirement ID	Priority	Requirement
FR-IAM-001	Must	The system shall allow user registration using a unique email address and password.
FR-IAM-002	Must	The system shall verify the user email address before enabling protected actions.
FR-IAM-003	Must	The system shall provide secure login, logout, password reset, and session timeout.
FR-IAM-004	Must	The system shall enforce account statuses including pending, active, suspended, deactivated, and archived.
FR-IAM-005	Must	The system shall route authenticated users to the appropriate role-based dashboard.
FR-IAM-006	Must	The system shall enforce RBAC and permission checks for web and API actions.
FR-IAM-007	Must	The system shall load and validate tenant context for tenant users.
FR-IAM-008	Must	The system shall log authentication, password, session, role, and permission events.
FR-IAM-009	Should	Users shall be able to view and revoke active sessions.
FR-IAM-010	Future	The architecture shall support OTP, social login, MFA, and SSO without major redesign.

8.2 Marketplace and Venue Discovery

Requirement ID	Priority	Requirement
FR-MKT-001	Must	Guests and players shall be able to browse published active venues.
FR-MKT-002	Must	Users shall be able to search by sport and location.
FR-MKT-003	Must	Users shall be able to filter by price, facilities, rating, distance, date, and availability.
FR-MKT-004	Must	Venue pages shall display business, location, grounds, sports, facilities, images, pricing guidance, policies, ratings, and availability.
FR-MKT-005	Must	Only published venues and operational grounds shall appear in bookable results.
FR-MKT-006	Should	Registered players shall be able to save and remove favorite venues.
FR-MKT-007	Should	Search results shall support sorting by relevance, distance, price, rating, and availability.
FR-MKT-008	Should	The system shall display map location and navigation information through a configured map provider.

8.3 Tenant, Venue and Ground Management

Requirement ID	Priority	Requirement
FR-TEN-001	Must	The system shall accept business applications and supporting documents.
FR-TEN-002	Must	Only approved businesses shall receive an active tenant workspace.
FR-TEN-003	Must	The system shall assign a unique tenant identifier and initialize default roles, settings, audit logs, and subscription data.
FR-VEN-001	Must	Tenant Admin shall be able to create, update, publish, archive, and manage multiple venues.
FR-GRD-001	Must	Each venue shall support multiple grounds or courts with independent sport, capacity, surface, lighting, equipment, and status.
FR-GRD-002	Must	Grounds shall support active, inactive, maintenance, and archived status.
FR-VEN-002	Must	The system shall validate required venue and ground information before publication.
FR-VEN-003	Should	Tenant Admin shall be able to manage facilities, images, contact information, policies, and operating hours.
FR-TEN-004	Must	Every tenant-owned record shall be associated with exactly one tenant identifier.
FR-TEN-005	Must	The system shall prevent cross-tenant reading, writing, exporting, and administrative actions.

8.4 Pricing and Availability

Requirement ID	Priority	Requirement
FR-PRI-001	Must	Tenant Admin shall be able to configure hourly pricing.
FR-PRI-002	Should	The pricing engine shall support peak, off-peak, weekend, holiday, seasonal, promotional, and membership pricing.
FR-PRI-003	Must	The system shall calculate the final booking price before payment and show taxes, fees, and discounts.
FR-AVL-001	Must	Tenant Admin shall configure operating hours and bookable slot duration.
FR-AVL-002	Must	Authorized tenant roles shall be able to block dates and time slots.
FR-AVL-003	Must	Ground maintenance status shall prevent new bookings for the affected period.
FR-AVL-004	Must	Availability shall synchronize with confirmed, pending, cancelled, expired, and maintenance events in real time.
FR-AVL-005	Must	The system shall revalidate availability immediately before confirmation.

8.5 Booking Engine

Requirement ID	Priority	Requirement
FR-BKG-001	Must	Only authenticated active players shall create customer bookings.
FR-BKG-002	Must	The booking engine shall validate user, tenant, subscription, venue, ground, date, slot, policy, price, and payment eligibility.
FR-BKG-003	Must	The system shall prevent double booking through transactional availability control.
FR-BKG-004	Must	The system shall create a unique booking identifier and maintain status history.
FR-BKG-005	Must	Paid bookings shall be confirmed only after verified payment.
FR-BKG-006	Should	The system shall support optional tenant-configured manual approval.
FR-BKG-007	Should	Eligible bookings shall support policy-based date, time, ground, player-count, or request modification.
FR-BKG-008	Must	Cancellation shall release the reserved slot and determine refund eligibility.
FR-BKG-009	Must	Authorized operational staff shall check customers in and update in-progress, completed, or no-show status.
FR-BKG-010	Must	All critical booking changes shall generate notifications and audit logs.

8.6 Payment, Invoice, Refund and Reconciliation

Requirement ID	Priority	Requirement
FR-PAY-001	Must	The system shall support secure payment initiation and gateway redirection or tokenized processing.
FR-PAY-002	Must	The system shall independently verify gateway callbacks before recording success.
FR-PAY-003	Must	Duplicate payment requests and duplicate callback processing shall be prevented.
FR-PAY-004	Must	Successful payment shall update the related booking or subscription atomically.
FR-PAY-005	Must	The system shall generate a receipt and invoice or transaction document after successful payment.
FR-PAY-006	Must	Refunds shall follow configured cancellation and tenant refund policies.
FR-PAY-007	Must	The system shall support full and partial refund statuses if supported by the selected provider.
FR-PAY-008	Must	Financial transactions shall be immutable after settlement except through controlled reversal or adjustment records.
FR-PAY-009	Should	The system shall reconcile platform records against gateway transactions and flag discrepancies.
FR-PAY-010	Should	The system shall support cards, wallets, internet banking, QR, and optional cash according to configured providers and tenant policy.

8.7 Tenant Staff and Field Administration

Requirement ID	Priority	Requirement
FR-STAFF-001	Must	Tenant Admin shall create, update, suspend, and remove tenant staff accounts.
FR-STAFF-002	Must	Tenant Admin shall assign predefined roles and permissions within the tenant.
FR-FIELD-001	Must	Field Admin shall access only assigned tenant venues and grounds.
FR-FIELD-002	Must	Field Admin shall view daily bookings, customer details required for service, and operational notes.
FR-FIELD-003	Must	Field Admin shall verify booking identity and check customers in.
FR-FIELD-004	Must	Field Admin shall update ground readiness and booking execution statuses.
FR-FIELD-005	Must	Field Admin shall record no-show, incident, maintenance, and operational exception details.
FR-FIELD-006	Must	Authorized Field Admin shall block affected slots and trigger notifications or escalation.
FR-FIELD-007	Must	Field Admin shall not manage tenant subscription, platform users, or unrestricted financial configuration.
FR-STAFF-003	Should	The tenant workspace should support staff scheduling and attendance records.

8.8 CRM, Reviews, Promotions and CMS

Requirement ID	Priority	Requirement
FR-CRM-001	Must	Tenant Admin shall view tenant-specific customer and booking history.
FR-CRM-002	Should	The system shall provide customer notes, repeat-booking information, and communication history subject to permissions.
FR-REV-001	Must	Only a player with an eligible completed booking shall submit a review for that booking.
FR-REV-002	Must	Tenant Admin shall be able to respond to reviews; Super Admin shall support moderation.
FR-PRO-001	Should	Tenant Admin shall create percentage, fixed, seasonal, first-booking, coupon, referral, membership, or event offers.
FR-PRO-002	Must	The booking engine shall validate promotion dates, eligibility, limits, and usage before applying a discount.
FR-CMS-001	Should	Authorized users shall manage news, events, gallery items, announcements, and static content.
FR-FEAT-001	Should	Tenants shall purchase or receive approved featured placement with configured duration and visibility.

8.9 Subscription Management

Requirement ID	Priority	Requirement
FR-SUB-001	Must	Super Admin shall create and configure subscription plans, limits, features, billing cycles, and status.
FR-SUB-002	Must	Tenant Admin shall view plans, current subscription, billing history, and renewal dates.
FR-SUB-003	Must	The system shall activate, renew, upgrade, downgrade, cancel, expire, suspend, and reactivate subscriptions.
FR-SUB-004	Must	Plan features and limits shall be enforced throughout the tenant workspace.
FR-SUB-005	Must	The system shall send renewal reminders before expiry.
FR-SUB-006	Must	Expired subscription access shall be restricted according to configured policy without deleting tenant data.
FR-SUB-007	Should	Plan changes shall calculate effective date, price differences, and billing adjustments.

8.10 Super Admin and Platform Governance

Requirement ID	Priority	Requirement
FR-ADM-001	Must	Super Admin shall review, approve, reject, request information for, suspend, reactivate, and terminate tenants.

Requirement ID	Priority	Requirement
FR-ADM-002	Must	Super Admin shall manage platform users, account statuses, and platform administrative roles.
FR-ADM-003	Must	Super Admin shall monitor subscription, booking, payment, refund, and revenue information platform-wide.
FR-ADM-004	Must	Super Admin shall configure sports categories, facilities, plans, platform settings, and content.
FR-ADM-005	Must	Super Admin shall search, filter, and export authorized audit and operational reports.
FR-ADM-006	Must	Super Admin shall monitor security events and perform controlled incident-response actions.
FR-ADM-007	Should	Super Admin shall manage advertisements, featured listings, and moderation queues.
FR-ADM-008	Should	Super Admin and support officers shall manage support cases and record resolutions.

8.11 Reporting, Analytics and Notifications

Requirement ID	Priority	Requirement
FR-RPT-001	Must	Tenant reports shall contain only tenant-owned data.
FR-RPT-002	Must	The system shall provide booking, revenue, occupancy, cancellation, refund, customer, staff, and rating metrics.
FR-RPT-003	Should	Authorized users shall filter and export reports in common formats.
FR-NTF-001	Must	The system shall notify users of verification, booking, payment, cancellation, refund, reminder, subscription, and security events.
FR-NTF-002	Must	Notification templates shall support email and in-app delivery; SMS may be enabled through the selected provider.
FR-NTF-003	Should	Users shall manage permitted notification preferences without disabling mandatory security or transaction messages.

9. Detailed End-to-End User Scenarios

The following scenarios define the expected system behaviour from the user perspective. They are implementation and test references. Policy-dependent values such as cancellation windows, approval rules, refund percentages, grace periods, and plan limits remain configurable.

9.1 Player and Guest User Scenarios

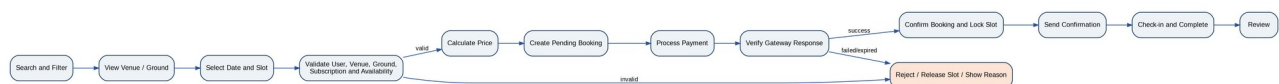


Figure 4. End-to-end customer booking flow

USR-01 - Browse and search venues

Primary actor	Guest User / Registered Player
Goal	Find suitable sports venues.
Preconditions	Published venue data is available.
Trigger	The actor opens the marketplace.
Main flow	1. Open the marketplace home page. 2. Select a sport or enter a location. 3. Apply available filters such as date, price, distance, facilities, rating, and availability. 4. The system validates search criteria and returns published active venues. 5. The actor sorts or refines the results. 6. The actor opens a venue detail page.
Alternate / exception flow	- If no results match, the system displays a clear empty state and allows criteria changes. - If the map service is unavailable, list results remain available without routing information.
Postconditions	Search activity is completed without exposing private tenant information.
Related requirements	FR-MKT-001 to FR-MKT-008

USR-02 - Register and verify account

Primary actor	Guest User
Goal	Create an active player account.
Preconditions	The email address is not already registered.
Trigger	The guest selects Register.
Main flow	1. Enter name, email, password, and required consent. 2. The system validates format and password policy. 3. The system checks for duplicate email. 4. The system creates a pending account. 5. A verification message is sent. 6. The user opens the verification link. 7. The system activates the account and records the event.
Alternate / exception flow	- Duplicate email: direct the user to login or password reset. - Expired verification link: allow a new link to be requested. - Invalid data: keep valid fields and display field-level errors.
Postconditions	A verified active player account exists.
Related requirements	FR-IAM-001 to FR-IAM-004

USR-03 - Login and recover password

Primary actor	Registered Player
Goal	Access the player dashboard securely.
Preconditions	The account exists.
Trigger	The player opens Login.
Main flow	1. Enter email and password. 2. The system validates credentials and account status. 3. The system creates a secure session. 4. The player is redirected to the customer dashboard. 5. For password recovery, the system sends a time-limited reset link to the verified email.
Alternate / exception flow	- Invalid credentials: display a generic error without revealing account existence. - Suspended or deactivated account: deny access and show the permitted support path. - Repeated failure: apply rate limiting or temporary lockout.
Postconditions	A secure session is created or a password-reset process is completed.
Related requirements	FR-IAM-003 to FR-IAM-009

USR-04 - View venue details and availability

Primary actor	Guest User / Registered Player
Goal	Evaluate a venue and available slots.
Preconditions	The venue is published and active.
Trigger	The actor selects a search result.
Main flow	1. Display venue business information, location, supported sports, facilities, images, ground details, policies, ratings, and pricing guidance. 2. The actor selects a ground and date. 3. The system loads current operating hours, blocks, maintenance, existing reservations, and subscription eligibility. 4. Only valid available slots are shown.
Alternate / exception flow	- If a ground is inactive or under maintenance, it is marked unavailable. - If availability changes during viewing, the system refreshes it before booking.
Postconditions	The actor can select a valid slot or return to search.
Related requirements	FR-MKT-004, FR-GRD-002, FR-AVL-001 to FR-AVL-005

USR-05 - Create and pay for a booking

Primary actor	Registered Player
Goal	Reserve an available ground successfully.
Preconditions	Player is authenticated and active; venue and tenant are eligible.
Trigger	The player selects an available slot.
Main flow	1. The system revalidates the player, tenant, venue, ground, date, slot, and policies. 2. The pricing engine calculates base price, discount, tax, fee, and total. 3. The player confirms booking details. 4. The system creates a pending booking and temporarily protects the slot. 5. The player completes payment through the configured provider. 6. The system verifies the payment response independently. 7. The system confirms the booking, locks the slot, records status history, creates invoice/receipt, and sends notifications.
Alternate / exception flow	- Validation failure: reject without payment and show the reason. - Payment failure or timeout: keep or release the slot according to configured timeout, then mark the booking failed or expired. - Duplicate callback: process idempotently without duplicate charge or confirmation.
Postconditions	A confirmed booking and successful payment record exist, or the attempt ends safely without double booking.
Related requirements	FR-BKG-001 to FR-BKG-010, FR-PAY-001 to FR-PAY-005

USR-06 - Modify an eligible booking

Primary actor	Registered Player
Goal	Change booking details within policy.
Preconditions	Booking is confirmed and still modifiable.
Trigger	The player selects Modify Booking.
Main flow	1. Display current booking and permitted changes. 2. The player selects a new date, slot, ground, player count, or request. 3. The system validates the policy and rechecks availability. 4. The system recalculates price difference. 5. The player confirms and completes additional payment or receives an adjustment path if required. 6. The system updates the booking, releases the old slot, reserves the new slot, logs the change, and sends notifications.
Alternate / exception flow	- If the new slot is no longer available, keep the original booking unchanged. - If modification is outside policy, deny the request and explain the restriction.
Postconditions	The booking is safely changed or remains unchanged.
Related requirements	FR-BKG-007, FR-AVL-004, FR-PAY-004, FR-NTF-001

USR-07 - Cancel booking and request refund

Primary actor	Registered Player
Goal	Cancel an eligible booking.
Preconditions	Booking exists and actor owns it.
Trigger	The player selects Cancel.
Main flow	1. Display cancellation policy and estimated refund. 2. The player confirms cancellation. 3. The system validates timing, booking status, payment, promotion, and tenant refund policy. 4. The system changes booking status, releases the slot, and creates an audit event. 5. If eligible, the system creates and processes a refund request. 6. The system updates refund and payment status and sends confirmation.
Alternate / exception flow	- Non-refundable booking: allow cancellation only after clear confirmation of zero refund. - Gateway refund failure: retain Refund Pending and alert authorized staff. - Already completed/no-show booking: deny normal cancellation.
Postconditions	The booking is cancelled, availability is restored, and refund status is traceable.
Related requirements	FR-BKG-008, FR-PAY-006 to FR-PAY-009

USR-08 - Attend, check in and complete booking

Primary actor	Registered Player and Field Admin
Goal	Deliver the reserved service.
Preconditions	Booking is confirmed and within the valid check-in window.
Trigger	The player arrives at the venue.
Main flow	1. The Field Admin searches or scans the booking identifier. 2. The system verifies booking, customer, venue, ground, date, time, and status. 3. The Field Admin checks the customer in. 4. The booking becomes Checked-in and then In Progress. 5. After the slot, authorized staff marks the booking Completed. 6. The system updates operational analytics and review eligibility.
Alternate / exception flow	- Early or late arrival: follow configured grace-period rules. - Invalid or cancelled booking: deny check-in and provide an escalation path. - Customer does not arrive: mark No-show according to policy.
Postconditions	The booking is completed or a documented exception status is recorded.
Related requirements	FR-FIELD-002 to FR-FIELD-005, FR-BKG-009

USR-09 - Submit rating and review

Primary actor	Registered Player
Goal	Provide feedback for a completed booking.
Preconditions	The player owns an eligible completed booking with no existing review.
Trigger	The player selects Rate and Review.
Main flow	1. Select rating and enter review text. 2. The system validates eligibility and content. 3. The review is saved with the moderation status. 4. The venue rating is recalculated according to the approved method. 5. Tenant Admin is notified and may respond.
Alternate / exception flow	- Ineligible booking or duplicate review: reject. - Reported or prohibited content: hold for moderation.
Postconditions	A traceable review exists and can be moderated or answered.
Related requirements	FR-REV-001 to FR-REV-002

USR-10 - Manage profile, favorites and notifications

Primary actor	Registered Player
Goal	Maintain account and preferences.
Preconditions	Player is authenticated.
Trigger	The player opens account settings.
Main flow	1. View and update permitted profile fields. 2. Add or remove favorite venues. 3. View booking, payment, and notification history. 4. Configure optional notification preferences. 5. Save changes and record security-sensitive updates.
Alternate / exception flow	- Email change requires re-verification. - Mandatory security and transaction notifications cannot be disabled.
Postconditions	Updated account preferences are stored securely.
Related requirements	FR-IAM-008, FR-MKT-006, FR-NTF-001 to FR-NTF-003

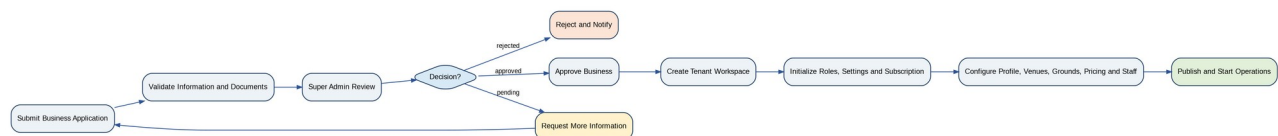
9.2 Tenant Admin / Company Admin Scenarios

Figure 5. Vendor application, verification and tenant provisioning flow

TAD-01 - Submit vendor application

Primary actor	Tenant Admin Applicant
Goal	Apply to join MeroMaidan as a sports business.
Preconditions	Applicant has required business information and documents.
Trigger	The applicant selects Register Business.
Main flow	1. Create or authenticate the owner account. 2. Enter business identity, registration number, contact, address, sports categories, PAN/VAT if applicable, and description. 3. Upload required documents using secure file controls. 4. Review declarations and submit. 5. The system validates mandatory fields, detects likely duplicates, creates a pending application, and notifies the applicant and Super Admin.
Alternate / exception flow	- Invalid or missing document: keep draft and show required correction. - Duplicate business: flag for manual review rather than creating an active tenant.
Postconditions	A traceable pending business application exists.
Related requirements	FR-TEN-001, FR-IAM-001 to FR-IAM-004

TAD-02 - Respond to verification request

Primary actor	Tenant Admin Applicant
Goal	Provide missing or corrected information.
Preconditions	Application is Pending Additional Information.
Trigger	The applicant receives a verification request.
Main flow	1. Open the secure application status page. 2. Review requested corrections. 3. Update permitted fields or replace documents. 4. Submit the response. 5. The system records version history and returns the application to review.
Alternate / exception flow	- Missed deadline: application remains pending or expires according to policy. - Rejected document type: show validation guidance.
Postconditions	The review team receives complete revised information.
Related requirements	FR-TEN-001, FR-ADM-001

TAD-03 - Configure initial tenant workspace

Primary actor	Tenant Admin
Goal	Prepare an approved tenant for operations.
Preconditions	Super Admin approved the business and workspace provisioning succeeded.
Trigger	Tenant Admin signs in for the first time.
Main flow	1. Accept tenant terms and confirm administrator profile. 2. Upload logo and cover image. 3. Configure business description, contacts, address, policies, and operating hours. 4. Review initialized plan and limits. 5. Create the first venue and ground. 6. Configure pricing and availability. 7. Invite required staff. 8. Complete readiness checklist and submit venues for publication.
Alternate / exception flow	- Incomplete mandatory setup: keep venue in Draft. - Subscription or plan limit prevents an action: explain the limit and available options.
Postconditions	The tenant workspace is configured and ready for publication.
Related requirements	FR-TEN-002 to FR-TEN-005, FR-VEN-001 to FR-VEN-003

TAD-04 - Manage venues and grounds

Primary actor	Tenant Admin
Goal	Maintain accurate venue and ground information.
Preconditions	Tenant is active.
Trigger	Tenant Admin opens Venue Management.
Main flow	1. Create or select a venue. 2. Enter or update address, coordinates, sports, facilities, images, contacts, policies, and schedule. 3. Create or update grounds with capacity, surface, lighting, equipment, classification, and status. 4. Validate mandatory data. 5. Publish, unpublish, deactivate, or archive according to permissions.
Alternate / exception flow	- Ground with future bookings cannot be deleted; use deactivation/archive and resolve bookings. - Publication fails when required fields are missing.
Postconditions	Marketplace and operational data reflect authorized changes.
Related requirements	FR-VEN-001 to FR-VEN-003, FR-GRD-001 to FR-GRD-002

TAD-05 - Configure pricing and availability

Primary actor	Tenant Admin
Goal	Define when and at what price grounds can be booked.
Preconditions	Venue and ground exist.
Trigger	Tenant Admin opens Pricing or Availability.
Main flow	1. Define operating hours and slot duration. 2. Create base hourly pricing. 3. Optionally add peak, off-peak, weekend, holiday, seasonal, promotional, or membership rules. 4. Block dates, holidays, or maintenance periods. 5. Preview resulting slots and prices. 6. Save and publish configuration.
Alternate / exception flow	- Overlapping pricing rules: apply documented priority and show a conflict warning. - Changes affecting existing bookings require explicit policy and confirmation.
Postconditions	Future availability and prices are generated consistently.
Related requirements	FR-PRI-001 to FR-PRI-003, FR-AVL-001 to FR-AVL-005

TAD-06 - Manage staff, roles and assignments

Primary actor	Tenant Admin
Goal	Provide controlled staff access.
Preconditions	Tenant is active and plan limits allow staff accounts.
Trigger	Tenant Admin opens Staff Management.
Main flow	1. Enter staff identity and contact information. 2. Select a predefined role such as Manager, Field Admin, Reception, Ground Supervisor, or Operations Staff. 3. Assign permitted venues or grounds. 4. Review effective permissions. 5. Send invitation or activate the account. 6. Later update, suspend, reassign, or remove the staff account.
Alternate / exception flow	- Duplicate email: link or invite according to membership rules. - Plan limit exceeded: prevent activation and present upgrade information. - Tenant Admin cannot grant platform-level or cross-tenant permissions.
Postconditions	Staff can access only authorized tenant resources.
Related requirements	FR-STAFF-001 to FR-STAFF-003, FR-FIELD-001, FR-IAM-006 to FR-IAM-008

TAD-07 - Monitor and manage bookings

Primary actor	Tenant Admin / Manager
Goal	Control tenant booking operations.
Preconditions	Tenant has venues and active bookings.
Trigger	Actor opens Booking Calendar or Booking List.
Main flow	1. Filter bookings by venue, ground, date, status, customer, or identifier. 2. Open a booking and view permitted customer, payment, and history information. 3. Approve or reject if manual approval is enabled. 4. Modify, cancel, refund, or resolve exceptions according to permissions and policy. 5. Record notes and communicate with the customer. 6. All actions update status, availability, notifications, and audit logs.
Alternate / exception flow	- Unauthorized financial action: deny and show permission requirement. - Status conflict: refresh current booking state before changes.
Postconditions	Booking operations remain synchronized and auditable.
Related requirements	FR-BKG-004 to FR-BKG-010, FR-PAY-006 to FR-PAY-009

TAD-08 - Manage customers, reviews and promotions

Primary actor	Tenant Admin / Authorized Manager
Goal	Improve customer engagement and bookings.
Preconditions	Tenant is active.
Trigger	Actor opens CRM, Reviews, or Promotions.
Main flow	1. View tenant-specific customer and booking history. 2. Add permitted internal notes. 3. Respond to eligible reviews. 4. Create a promotion with type, value, dates, limits, eligible venues/grounds, and customer conditions. 5. Preview and activate the promotion. 6. Monitor usage and performance.
Alternate / exception flow	- Promotion conflicts or exceeds policy: prevent activation. - Review violates moderation rules: escalate to Super Admin rather than delete without authority.
Postconditions	Customer activity and promotions are managed within tenant scope.
Related requirements	FR-CRM-001 to FR-CRM-002, FR-REV-002, FR-PRO-001 to FR-PRO-002

TAD-09 - View reports and analytics

Primary actor	Tenant Admin / Manager
Goal	Understand tenant performance.
Preconditions	Tenant has operational data.
Trigger	Actor opens Analytics.
Main flow	1. Select date range, venues, grounds, sports, and report type. 2. The system retrieves tenant-scoped data only. 3. Display bookings, revenue, occupancy, payment success, cancellation, no-show, customer, rating, and staff metrics. 4. Compare periods where supported. 5. Export authorized reports.
Alternate / exception flow	- No data: show a clear empty state. - Large export: process asynchronously and notify when ready.
Postconditions	A tenant-scoped report or export is available.
Related requirements	FR-RPT-001 to FR-RPT-003

TAD-10 - Manage subscription and billing

Primary actor	Tenant Admin
Goal	Maintain access to tenant services.
Preconditions	Tenant is approved and has subscription data.
Trigger	Tenant Admin opens Subscription.
Main flow	1. View current plan, features, limits, renewal date, and billing history. 2. Compare available plans. 3. Request upgrade, downgrade, renewal, or cancellation. 4. The system calculates pricing and effective date. 5. Tenant Admin completes payment if required. 6. The system verifies payment, updates subscription, applies limits, issues receipt, and sends notification.
Alternate / exception flow	- Failed payment: do not activate paid changes. - Downgrade below current resource usage: require remediation or scheduled enforcement. - Expired plan: restrict configured features while preserving data.
Postconditions	Subscription status and access rules are updated consistently.
Related requirements	FR-SUB-001 to FR-SUB-007, FR-PAY-001 to FR-PAY-005

9.3 Field Admin and Field Operations Scenarios

Figure 6. Field Admin daily operational flow

FAD-01 - Access assigned field dashboard

Primary actor	Field Admin
Goal	View operational information only for assigned locations.
Preconditions	Field Admin account is active and assigned to tenant venues or grounds.
Trigger	Field Admin logs in.
Main flow	1. The system authenticates the user and loads tenant context. 2. The system loads assigned venues, grounds, date, and operational permissions. 3. Dashboard displays current and upcoming bookings, ground status, pending check-ins, maintenance blocks, and alerts. 4. Financial and subscription functions remain hidden or denied.
Alternate / exception flow	- No active assignment: display a restricted dashboard and instruct the user to contact Tenant Admin. - Cross-tenant or unassigned URL request: deny and log.
Postconditions	Field Admin sees only assigned operational data.
Related requirements	FR-FIELD-001, FR-FIELD-007, FR-IAM-006 to FR-IAM-008

FAD-02 - Review daily schedule and ground readiness

Primary actor	Field Admin
Goal	Prepare assigned grounds before bookings.
Preconditions	Assigned grounds and daily schedule exist.
Trigger	Field Admin starts a shift.
Main flow	1. Open the daily schedule. 2. Review bookings, setup notes, equipment, and maintenance warnings. 3. Inspect each ground. 4. Mark Ready, Attention Required, or Unavailable. 5. If unavailable, provide reason, expected duration, and evidence if required. 6. The system updates operational status and alerts authorized tenant users.
Alternate / exception flow	- Unexpected closure affecting bookings: start incident and slot-block process. - Missing equipment: record issue and escalate.
Postconditions	Ground readiness is visible to authorized staff and booking controls.
Related requirements	FR-FIELD-002, FR-FIELD-004 to FR-FIELD-006

FAD-03 - Verify booking and check in customer

Primary actor	Field Admin / Reception Staff
Goal	Admit the correct customer to the correct booking.
Preconditions	Booking is confirmed and within check-in policy.
Trigger	Customer presents booking reference.
Main flow	1. Search or scan the booking reference. 2. Verify customer name, venue, ground, date, time, status, and permitted payment indicator. 3. Confirm identity using the approved method. 4. Record arrival time and check the customer in. 5. The system updates status and notifies relevant operational users if configured.
Alternate / exception flow	- Booking not found or invalid: deny check-in and escalate. - Customer arrives outside grace period: apply policy and record outcome. - Payment status unresolved: refer to authorized manager rather than overriding without permission.
Postconditions	A valid booking becomes Checked-in or a documented exception is recorded.
Related requirements	FR-FIELD-002 to FR-FIELD-005

FAD-04 - Start, complete or mark no-show

Primary actor	Field Admin
Goal	Record actual booking execution.
Preconditions	Booking is confirmed or checked in.
Trigger	Scheduled start or end time is reached.
Main flow	1. For an arrived customer, mark In Progress when use begins. 2. At the end, mark Completed and record permitted notes or incidents. 3. For a customer who does not arrive within the grace period, mark No-show. 4. The system updates status history, availability, customer history, and analytics.
Alternate / exception flow	- Booking must not skip prohibited state transitions. - Late operational completion must not alter original booked time without an authorized adjustment.
Postconditions	The booking has an accurate final operational status.
Related requirements	FR-BKG-009, FR-FIELD-004 to FR-FIELD-005

FAD-05 - Block a slot for maintenance or safety

Primary actor	Field Admin / Ground Supervisor
Goal	Prevent unsafe or unavailable ground use.
Preconditions	Actor has block permission for the assigned ground.
Trigger	A maintenance or safety issue is identified.
Main flow	1. Open the ground schedule. 2. Select the affected date and time range. 3. Enter issue category, reason, severity, expected resolution, and notes. 4. The system checks affected bookings. 5. Confirm the block and create an incident or maintenance record. 6. The system prevents new bookings and notifies Tenant Admin and affected customers according to policy.
Alternate / exception flow	- Existing confirmed booking affected: require escalation and controlled cancellation, relocation, or refund flow. - Actor lacks permission for a long-term block: save incident and request approval.
Postconditions	Affected slots are unavailable and the issue is traceable.
Related requirements	FR-AVL-002 to FR-AVL-004, FR-FIELD-005 to FR-FIELD-006

FAD-06 - Record and escalate an operational incident

Primary actor	Field Admin
Goal	Create a traceable operational record.
Preconditions	Actor is assigned to the location.
Trigger	An injury, dispute, equipment failure, security event, or service issue occurs.
Main flow	1. Open Incident Report. 2. Select booking, venue, ground, category, severity, and time. 3. Enter factual description and immediate action. 4. Attach permitted evidence. 5. Select escalation recipient. 6. Submit; the system timestamps, audits, and notifies authorized users. 7. Update the record until resolved according to permissions.
Alternate / exception flow	- Critical security event: immediately alert Super Admin/security process in addition to tenant management. - Sensitive evidence: restrict access to authorized roles.
Postconditions	An auditable incident exists with status and ownership.
Related requirements	FR-FIELD-005 to FR-FIELD-006, FR-ADM-006, FR-IAM-008

FAD-07 - Complete end-of-day operational summary

Primary actor	Field Admin / Manager
Goal	Close the shift with accurate operational information.
Preconditions	Shift or operating day is ending.
Trigger	Actor opens Daily Summary.
Main flow	1. Review completed, cancelled, no-show, and unresolved bookings. 2. Review ground status, incidents, maintenance, and outstanding tasks. 3. Correct permitted incomplete operational statuses. 4. Enter handover notes. 5. Submit the summary. 6. The system records the submission and makes it available to authorized tenant management.
Alternate / exception flow	- Unresolved critical issue: summary cannot be closed without owner and next action. - Missing booking status: prompt actor to complete or escalate.
Postconditions	The tenant has a traceable operational handover record.
Related requirements	FR-FIELD-002 to FR-FIELD-006, FR-RPT-001

9.4 Super Admin and Platform Governance Scenarios**SAD-01 - Review and decide vendor application**

Primary actor	Super Admin
Goal	Approve only legitimate and complete businesses.
Preconditions	A vendor application is pending.
Trigger	Super Admin opens the verification queue.
Main flow	1. Review business information, documents, duplicate warnings, and verification results. 2. Record verification notes. 3. Choose Approve, Reject, or Request Additional Information. 4. For approval, confirm initial plan and administrator identity. 5. The system records the decision and notifies the applicant. 6. For approval, the system provisions the tenant workspace.
Alternate / exception flow	- Suspicious or incomplete application: keep pending and request evidence. - Provisioning failure: keep Approved-Pending-Provisioning and alert technical operations.
Postconditions	A documented decision exists and approved tenants receive an isolated workspace.
Related requirements	FR-ADM-001, FR-TEN-002 to FR-TEN-003

SAD-02 - Manage tenant lifecycle

Primary actor	Super Admin
Goal	Govern tenant status safely.
Preconditions	Tenant exists.
Trigger	Super Admin opens Tenant Management.
Main flow	1. View tenant profile, status, subscription, violations, security events, and audit history. 2. Select suspend, reactivate, terminate, or archive according to policy. 3. Enter reason, effective date, access impact, and required notifications. 4. Confirm the high-risk action. 5. The system applies restrictions, preserves data, logs the action, and notifies authorized tenant users.
Alternate / exception flow	- Active bookings exist: require an operational resolution plan. - Termination: archive data according to retention policy; do not immediately hard-delete records.
Postconditions	Tenant status and access reflect the approved lifecycle decision.
Related requirements	FR-ADM-001, FR-SUB-006, FR-TEN-005

SAD-03 - Manage platform users and roles

Primary actor	Super Admin
Goal	Maintain platform administrative access.
Preconditions	Super Admin is authorized for user administration.
Trigger	Super Admin opens User Management.
Main flow	1. Search users by identity, role, tenant, status, or risk indicator. 2. View account status, memberships, sessions, and permitted audit information. 3. Suspend, reactivate, reset access, or assign platform administrative roles. 4. Confirm privileged changes. 5. The system applies the change and records old/new values in audit logs.
Alternate / exception flow	- Tenant staff role changes should normally be handled by Tenant Admin unless platform intervention is justified. - Super Admin cannot remove the final required platform administrator without a controlled process.
Postconditions	Platform user access is updated and auditable.
Related requirements	FR-ADM-002, FR-IAM-004 to FR-IAM-009

SAD-04 - Configure subscription plans and platform catalogue

Primary actor	Super Admin
Goal	Control sellable SaaS plans and reference data.
Preconditions	Super Admin has configuration permission.
Trigger	Super Admin opens Platform Configuration.
Main flow	1. Create or update plan name, description, price, billing cycle, trial, limits, features, and status. 2. Configure sports categories, facilities, standard statuses, and platform content settings. 3. Validate dependencies and preview impact. 4. Publish the configuration with effective date. 5. The system versions the configuration and logs the change.
Alternate / exception flow	- Reducing a limit below current tenant usage: require a migration or grandfathering rule. - Disabling a category with active venues: block immediate removal and use deactivation.
Postconditions	New configuration is applied consistently and traceably.
Related requirements	FR-SUB-001, FR-ADM-004

SAD-05 - Monitor payments and reconcile transactions

Primary actor	Super Admin / Finance-authorized Admin
Goal	Ensure financial records match providers.
Preconditions	Payment data and gateway settlement data exist.
Trigger	Actor opens Financial Operations.
Main flow	1. Filter transactions by date, tenant, booking, subscription, status, provider, or amount. 2. Retrieve or import gateway settlement information. 3. Match platform and gateway records. 4. Flag missing, duplicate, mismatched, or unresolved entries. 5. Assign and resolve reconciliation cases. 6. Export the authorized reconciliation report.
Alternate / exception flow	- Unverified callback or suspicious transaction: do not manually mark successful without evidence and authorized process. - Gateway unavailable: defer reconciliation and preserve the case.
Postconditions	Financial discrepancies are identified and resolved with audit history.
Related requirements	FR-PAY-002 to FR-PAY-009, FR-ADM-003

SAD-06 - Manage content, reviews and featured listings

Primary actor	Super Admin
Goal	Maintain platform quality and visibility controls.
Preconditions	Content or moderation item exists.
Trigger	Super Admin opens CMS or Moderation.
Main flow	1. Create or update platform news, announcements, banners, and static content. 2. Review reported reviews or tenant responses. 3. Approve, hide, reject, or restore content according to policy. 4. Manage featured listing eligibility, payment status, placement, duration, and expiry. 5. Publish and audit the action.
Alternate / exception flow	- Paid featured listing without verified payment: do not activate. - Content under dispute: preserve original record and moderation history.
Postconditions	Public content and promotional placement reflect approved decisions.
Related requirements	FR-CMS-001, FR-FEAT-001, FR-ADM-007

SAD-07 - Monitor security and respond to incident

Primary actor	Super Admin / Security-authorized Admin
Goal	Protect accounts, tenants, and platform integrity.
Preconditions	A security alert or suspicious event exists.
Trigger	The monitoring system or user raises an alert.
Main flow	1. Review event details, affected user/tenant, IP, device, module, action, status, and risk level. 2. Classify severity and scope. 3. Contain by revoking sessions, disabling tokens, suspending accounts, or restricting access as authorized. 4. Investigate related audit and operational events. 5. Record resolution and recovery actions. 6. Complete a post-incident review and required notifications.
Alternate / exception flow	- Potential cross-tenant leakage: treat as critical and preserve evidence. - False positive: document rationale before closure.
Postconditions	Incident actions, evidence, impact, and resolution are auditable.
Related requirements	FR-ADM-006, FR-IAM-008, NFR-SEC requirements

10. Business Rules and Status Models**10.1 Core Business Rules**

Rule ID	Business rule
BR-001	Every tenant, user, venue, ground, booking, payment, and subscription shall have a unique identifier.
BR-002	Every tenant-owned business record shall belong to exactly one tenant.

Rule ID	Business rule
BR-003	Cross-tenant data access is prohibited except for explicitly authorized platform administration.
BR-004	Only approved active tenants may publish venues and receive new bookings.
BR-005	Only published active venues and operational grounds may be booked.
BR-006	Only available time slots may be booked; double booking is prohibited.
BR-007	Availability shall be revalidated immediately before confirmation.
BR-008	Paid bookings require independently verified successful payment.
BR-009	Cancelled or expired bookings shall release protected slots according to the booking-lock policy.
BR-010	Refund eligibility shall follow configured tenant and platform policies.
BR-011	Pricing changes apply to future bookings unless an authorized migration policy states otherwise.
BR-012	Grounds under maintenance shall not accept new bookings for affected periods.
BR-013	Only completed eligible bookings may receive one player review.
BR-014	Subscription status and plan limits determine access to premium tenant features.
BR-015	Field Admin may operate only assigned venues/grounds and may not administer the tenant subscription or platform.
BR-016	All privileged, financial, booking-status, permission, tenant-status, and configuration changes shall be audited.
BR-017	Settled financial records and completed booking history shall not be destructively edited.
BR-018	Tenant termination shall archive records according to approved retention policy.

10.2 Booking Status Model

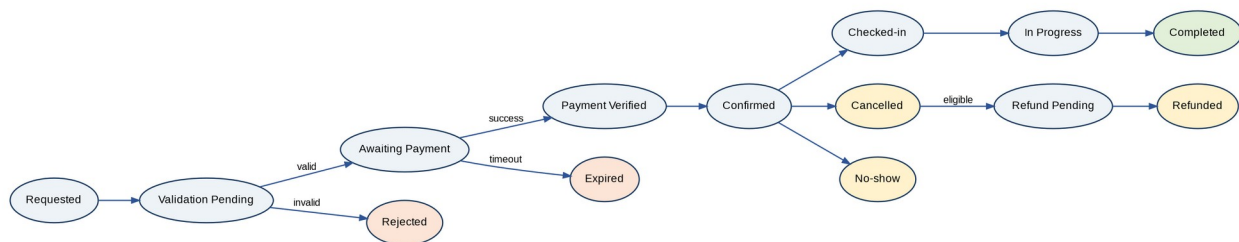


Figure 7. Booking state model

Status	Meaning
Draft	Booking details are being prepared but have not been submitted.
Requested	Booking request has been submitted.
Validation Pending	System validations are running.
Awaiting Payment	Booking is temporarily pending payment.
Payment Verified	Payment success has been independently confirmed.

Status	Meaning
Confirmed	The venue slot is reserved.
Checked-in	Customer arrival has been verified.
In Progress	The customer is using the ground.
Completed	Service was delivered successfully.
Cancelled	Booking was cancelled by an authorized actor or rule.
Refund Pending	A refund request is being processed.
Refunded	Refund completed.
Rejected	Booking failed validation or approval.
Expired	Pending booking or payment timed out.
No-show	Customer failed to attend within policy.

10.3 Tenant and Subscription Statuses

Object	Statuses
Tenant application	Draft, Submitted, Under Review, Additional Information Required, Approved, Rejected.
Tenant workspace	Provisioning, Active, Suspended, Reactivation Pending, Terminated, Archived.
Subscription	Trial, Active, Renewal Due, Past Due, Expired, Suspended, Cancelled.
Ground	Draft, Active, Inactive, Maintenance, Archived.
Payment	Pending, Processing, Successful, Failed, Cancelled, Expired, Partially Refunded, Refunded.
Incident	Open, Investigating, Action Required, Resolved, Closed.

11. Data Requirements and Logical Model

The following logical model is a concise implementation recommendation based on the business requirements. Detailed physical schema, indexes, constraints, and migrations shall be defined in the Software Design and Database Design documents.

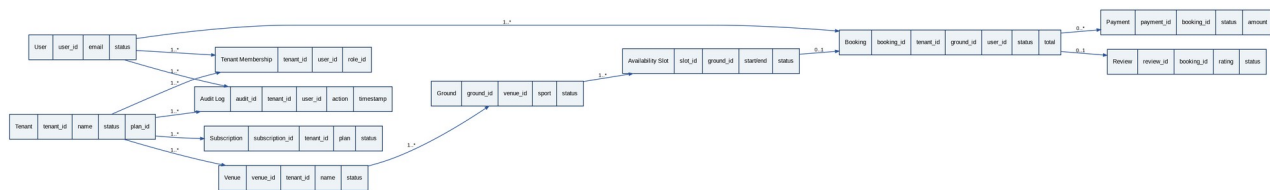


Figure 8. Simplified logical data model

Entity group	Purpose
Tenant	Business identity, verification status, subscription context, settings, and ownership boundary.
User	Identity, contact, verification, status, password/security information, and profile.
Role / Permission / Tenant Membership	Role definitions, granular permissions, user-to-tenant membership, and assignments.

Entity group	Purpose
Venue	Tenant-owned venue identity, location, profile, facilities, images, policies, and publication status.
Ground	Venue-owned bookable unit including sport, capacity, surface, equipment, status, and assignment.
Operating Hour / Availability Slot / Block	Normal schedules, generated slots, holidays, manual blocks, and maintenance periods.
Pricing Rule	Base and conditional prices, discount rule references, effective dates, and priority.
Booking / Booking Status History	Customer reservation, selected slot, calculated amount, status, policy snapshot, and immutable history.
Payment / Refund / Invoice	Gateway reference, amount, currency, status, verification, settlement, refund, and documents.
Subscription Plan / Subscription / Billing History	Plan features, limits, price, cycle, tenant subscription lifecycle, and billing events.
Customer Profile / Favorite / Review	Customer preferences, tenant relationship, saved venues, rating, text, moderation, and response.
Staff Assignment / Attendance	Tenant staff role, assigned venue/ground, schedule, attendance, and status.
Maintenance / Incident	Operational issue, severity, evidence, affected resources, actions, ownership, and resolution.
Promotion / Coupon / Featured Listing	Eligibility, value, limits, usage, payment, placement, and validity.
Notification	Recipient, channel, template, event, delivery status, retries, and timestamps.
Audit Log	Actor, tenant, module, action, resource, previous/new values, IP/device, status, and risk level.
Support Ticket / CMS Content	Support workflow and platform/tenant-managed public content.

11.1 Data Integrity Requirements

- Use foreign keys and application policies to preserve tenant, venue, ground, booking, payment, and subscription relationships.
- Use unique constraints for identifiers, emails where applicable, gateway references, invoice numbers, and idempotency keys.
- Use transactions and locking or equivalent concurrency controls for slot reservation and payment confirmation.
- Retain status history instead of overwriting the only evidence of a lifecycle transition.
- Use soft deletion or archival for business records that must remain traceable.
- Store timestamps in a consistent server standard and display them in the applicable local timezone.
- Separate sensitive credentials and secrets from normal application data; never store passwords or gateway secrets in plain text.
- Back up database and file storage according to the approved recovery objectives.

12. API and External Integration Requirements

12.1 API Standards

ID	Requirement
API-001	All protected endpoints shall require authentication and authorization.
API-002	Tenant endpoints shall derive and validate the active tenant context; client-supplied tenant IDs shall never be trusted alone.
API-003	APIs shall use versioned HTTPS endpoints and JSON request/response bodies.
API-004	Input shall be validated server-side and errors shall use consistent status codes and machine-readable messages.

ID	Requirement
API-005	List endpoints shall support pagination, filtering, sorting, and authorized search.
API-006	Payment and other retryable commands shall use idempotency controls.
API-007	Sensitive endpoints shall use rate limiting and abuse monitoring.
API-008	API actions shall produce audit events where required.
API-009	API documentation shall define request fields, validation, permissions, responses, errors, and examples.
API-010	Tokens and credentials shall support expiration and revocation.

12.2 Primary API Resource Groups

Resource group	Representative operations
Authentication	Register, verify email, login, logout, reset password, sessions.
Marketplace	Sports, locations, venue search, venue details, grounds, availability, reviews.
Bookings	Create, validate, view, modify, cancel, approve, check-in, complete, no-show, history.
Payments	Initiate, verify, callback/webhook, receipt, invoice, refund, reconciliation.
Tenant	Application, status, workspace, business profile, settings, subscription.
Venues and grounds	Create, update, publish, archive, facilities, images, operating hours, blocks, maintenance.
Staff and field operations	Invitations, roles, assignments, daily schedule, readiness, check-in, incidents, handover.
CRM and marketing	Customers, notes, reviews, responses, promotions, coupons, featured listings.
Reports and analytics	Tenant reports, platform reports, KPIs, exports.
Super Admin	Tenants, users, plans, catalogue, moderation, CMS, security, audit, support.

12.3 External Integrations

Integration	Requirements
Payment gateway	Secure initiation, signed/verified callback, idempotency, transaction query, refunds, reconciliation, test environment, and failure handling.
Email service	Verification, reset, transaction, booking, refund, subscription, and security messages with retry and delivery status.
SMS service	Optional OTP and high-value notifications based on cost and provider selection.
Map/location service	Geocoding, coordinates, distance/radius search, map display, and navigation links.
File storage	Secure upload, type/size validation, malware scanning where available, access control, and lifecycle management.

13. User Interface and Experience Requirements

13.1 General UX Requirements

- Responsive layouts shall support desktop, tablet, and mobile web use.
- Navigation, terminology, buttons, statuses, forms, and alerts shall remain consistent across portals.
- Each role shall see only relevant modules and actions; hidden navigation does not replace backend authorization.
- Forms shall show required fields, validation messages, safe defaults, progress, and clear success/failure states.
- Booking and payment screens shall show venue, ground, date, time, price breakdown, policy, and final confirmation before commitment.
- Field Admin pages shall prioritize speed, large touch targets, daily schedule, booking search, check-in, and ground status.
- Destructive or high-risk actions shall require confirmation and show impact.
- Empty, loading, offline, timeout, and error states shall provide useful next actions.
- Accessibility should follow recognized web accessibility practices, including keyboard navigation, labels, contrast, focus, and screen-reader-friendly structure.

13.2 Required Screens by Portal

Portal	Required screens
Public / Player	Home, search results, map/list, venue detail, ground/availability, register/login, booking summary, payment, confirmation, dashboard, bookings, booking detail, favorites, reviews, profile, notifications, support.
Tenant Admin	Dashboard, business profile, venues, grounds, pricing, availability/calendar, bookings, customers/CRM, staff/roles, field operations, ERP/resources, promotions, CMS, reports, analytics, subscription/billing, settings, audit.
Field Admin	Assigned dashboard, daily schedule, booking search/detail, check-in, ground readiness, status update, maintenance/block, incident, end-of-day summary, notifications.
Super Admin	Dashboard, verification queue, tenants, users, plans, payments, reconciliation, featured listings, advertisements, CMS, moderation, reports, security events, audit logs, support, platform configuration.

14. Non-Functional Requirements

ID	Category	Priority	Requirement
NFR-SEC-001	Security	Must	All application traffic shall use HTTPS.
NFR-SEC-002	Security	Must	Passwords shall be stored using a current secure hashing mechanism supported by Laravel.
NFR-SEC-003	Security	Must	The system shall implement server-side validation, output encoding, CSRF protection, secure cookies, security headers, ORM/parameterized queries, and safe file uploads.
NFR-SEC-004	Security	Must	Authorization and tenant ownership shall be verified on every protected resource operation.
NFR-PERF-001	Performance	Must	Authentication should complete within 2 seconds under normal operating conditions.

ID	Category	Priority	Requirement
NFR-PERF-002	Performance	Must	Authorization checks should normally complete within 100 milliseconds excluding network latency and external dependencies.
NFR-PERF-003	Performance	Must	Common pages and API responses should meet an agreed performance budget under expected load.
NFR-PERF-004	Performance	Must	Booking availability and confirmation shall use concurrency-safe processing to prevent conflicting reservations.
NFR-AVL-001	Availability	Should	The platform should target 99.9 percent service availability excluding approved maintenance.
NFR-AVL-002	Availability	Must	Planned maintenance shall be communicated and critical services shall fail gracefully.
NFR-SCAL-001	Scalability	Must	The architecture shall support thousands of users, hundreds of tenants, and high concurrent booking volume without redesign.
NFR-SCAL-002	Scalability	Should	Application services and queue workers should support horizontal scaling.
NFR-DATA-001	Backup	Must	Automated database and file backups shall be monitored and periodically restored in tests.
NFR-DATA-002	Recovery	Must	Recovery point and recovery time objectives shall be approved before production.
NFR-OBS-001	Observability	Must	The platform shall centralize application logs, security events, job failures, integration errors, and performance metrics.
NFR-OBS-002	Observability	Must	Critical alerts shall identify severity, affected tenant or service where applicable, and response owner.
NFR-MNT-001	Maintainability	Must	The codebase shall be modular, documented, version controlled, reviewed, and tested.
NFR-MNT-002	Maintainability	Must	Configuration and secrets shall be environment-specific and not hard-coded in source.
NFR-COMP-001	Compatibility	Must	The responsive web application shall support current major desktop and mobile browsers agreed for release.
NFR-UX-001	Usability	Must	Core booking and field-operation tasks shall be understandable without specialist training.
NFR-ACC-001	Accessibility	Should	Public and authenticated web interfaces should align with recognized accessibility guidelines.
NFR-LOC-001	Localization	Should	The design shall support future multilingual content and local date, time, currency, and address formats.

ID	Category	Priority	Requirement
NFR-PRIV-001	Privacy	Must	Personal data collection and use shall be limited to legitimate product and operational purposes.
NFR-AUD-001	Auditability	Must	Audit records shall be protected from unauthorized modification and retained according to policy.

15. Security, Privacy, Audit and Compliance

15.1 Security Controls

Control area	Required controls
Authentication	Verified accounts, strong password policy, secure reset, rate limiting, session regeneration, inactivity timeout, session revocation, and optional future MFA.
Authorization	RBAC, permissions, policy checks, tenant context, assigned-resource checks, least privilege, separation of duties, and denial by default.
Application security	HTTPS, CSRF, XSS prevention, SQL injection prevention, input validation, output encoding, secure headers, dependency updates, and file validation.
API security	Token expiration/revocation, rate limiting, standardized errors, signature verification, idempotency, and tenant-aware validation.
Data security	Encryption for sensitive data where applicable, protected backups, restricted secrets, data minimization, retention, and secure disposal.
Operational security	Central logs, alerts, vulnerability review, incident response, backup testing, environment separation, and least-privilege infrastructure access.

15.2 Audit Log Requirements

- Capture audit ID, timestamp, actor user ID, tenant ID when applicable, role, module, action, resource, status, IP address, device information, and risk level.
- Capture previous and new values for sensitive configuration or status changes where lawful and practical.
- Log authentication, user, role, permission, tenant, venue, booking, payment, refund, subscription, configuration, API-token, moderation, and security events.
- Authorized users shall search, filter, and export audit information within their permitted scope.
- Audit records shall not expose secrets, full payment credentials, or unnecessary sensitive personal data.

15.3 Incident Response

1. Detect and log the event.
2. Assess risk, scope, and affected users or tenants.
3. Classify severity and assign an owner.
4. Contain the threat and preserve evidence.
5. Investigate root cause and impact.
6. Resolve and recover services safely.
7. Notify affected stakeholders or authorities when required by policy or law.
8. Complete post-incident review and corrective actions.

15.4 Compliance and Privacy

The platform shall be designed to support applicable Nepalese electronic-transaction, privacy, consumer, tax, and business requirements. The uploaded sources do not define specific legal retention periods or regulatory interpretations; these require qualified legal and financial confirmation before production.

16. Reporting, Analytics and Notifications

16.1 Tenant Reports

- Daily, weekly, monthly, and annual booking and revenue reports.
- Venue and ground occupancy and utilization.
- Peak hours, sport-category performance, and booking trends.
- Payment success, failed payments, refunds, and outstanding transactions.
- Cancellation, modification, no-show, and completion rates.
- Customer growth, repeat customers, retention, and ratings.
- Promotion and coupon performance.
- Staff activity, check-ins, incidents, and operational completion where permitted.

16.2 Platform Reports

- Active users and tenants.
- Vendor application and activation funnel.
- Subscription plans, renewals, expiries, retention, and monthly recurring revenue.
- Marketplace bookings, gross value, platform revenue, and tenant revenue views.
- Payment reconciliation and refund status.
- Featured listing and advertising performance.
- Support resolution and security events.
- System availability, errors, queue health, and integration status.

16.3 Notification Events

Audience	Events
Player	Email verification, password reset, booking created/confirmed/rejected/modified/cancelled, payment result, refund, reminders, check-in, completion, review request, support, and security.
Tenant Admin	Application status, workspace provisioning, booking events, payment/refund exceptions, staff invitation, incidents, subscription reminders, plan changes, and security.
Field Admin	Assignments, schedule changes, new/changed booking, check-in reminder, ground issue, maintenance, incident escalation, and handover.
Super Admin	Vendor applications, provisioning failure, payment/reconciliation exception, critical incident, suspicious access, integration failure, support escalation, and system alert.

17. Testing Strategy and Acceptance Criteria

17.1 Required Test Levels

Test level	Coverage
Unit testing	Business services, pricing, validation, status transitions, permissions, and helper functions.
Integration testing	Database transactions, payment provider, email/SMS, maps, storage, queues, and scheduled jobs.
API testing	Authentication, authorization, tenant isolation, validation, errors, idempotency, pagination, and rate limits.
End-to-end testing	Customer booking, vendor onboarding, tenant setup, field operations, subscription, refund, and Super Admin workflows.
Security testing	Access control, cross-tenant attempts, session handling, common web vulnerabilities, file upload, secrets, and dependency scanning.

Test level	Coverage
Performance testing	Search, availability, booking concurrency, payment callback, dashboards, reports, and queue throughput.
Backup/recovery testing	Database and file restore, documented recovery steps, and data integrity after recovery.
User acceptance testing	Player, Tenant Admin, Field Admin, staff, Super Admin, and business-policy validation.

17.2 Release Acceptance Criteria

ID	Acceptance criterion
AC-001	A verified player can search, view live availability, book, pay, receive confirmation, attend, and review without manual platform intervention.
AC-002	Concurrent attempts cannot confirm the same ground and time slot.
AC-003	Failed, expired, or duplicate payment events do not create duplicate bookings or charges.
AC-004	An approved vendor receives a tenant workspace and can configure business profile, venues, grounds, pricing, availability, staff, and subscription.
AC-005	Tenant users cannot access another tenant through UI, API, identifier changes, exports, or direct URL requests.
AC-006	Field Admin can perform assigned operational tasks and cannot access prohibited tenant or platform functions.
AC-007	Booking cancellation releases the slot and the refund status follows configured policy.
AC-008	Super Admin can review vendors, govern tenants, manage plans, monitor payments, and inspect security/audit information.
AC-009	Required notifications and audit events are generated for critical lifecycles.
AC-010	Reports accurately reflect source transactions and respect role and tenant scope.
AC-011	Backup restore and critical incident procedures are demonstrated before production.
AC-012	No unresolved critical or high-severity security defects remain at release approval.

17.3 Definition of Done for a Feature

- Approved requirement and acceptance criteria implemented.
- Frontend, backend, database, API, permissions, and audit impact completed.
- Automated tests added and passing.
- Error, loading, empty, validation, and unauthorized states handled.
- Security and tenant-isolation review passed.
- Documentation and configuration updated.
- User acceptance completed for relevant role.
- No unresolved critical defects.

18. Assumptions, Dependencies and Risks

18.1 Assumptions and Decisions Required

Item	Current position / required decision
Payment provider	The platform will integrate with a Nepal-market provider, but the selected provider and capabilities are not confirmed.
Subscription plans	Plan names, prices, limits, free/trial policy, and commission model require business approval.
Booking approval	Automatic confirmation is the default after validation/payment; tenants may optionally enable manual approval.
Cancellation/refund	Windows, fees, percentages, grace periods, and exceptional overrides must be configurable and approved.
Cash payment	Optional and tenant-policy dependent; operational and reconciliation rules require confirmation.
Field Admin role	Included as a tenant-scoped operational role requested by the user, aligned with Manager/Ground Supervisor responsibilities.
Frontend	Next.js or React.js is recommended in the source; final framework requires technical confirmation.
Legal retention	Data, financial, audit, and incident retention periods require legal/financial approval.
SMS and maps	Provider selection and cost model remain dependencies.
Deployment	Production hosting model, recovery objectives, monitoring stack, and support model require technical approval.

18.2 Key Risks and Mitigations

Risk	Impact	Mitigation
Cross-tenant data leakage	Critical	Tenant context middleware, ownership policies, scoped queries, automated isolation tests, audit, and security review.
Double booking	Critical	Transactional slot reservation, locking/unique constraints, revalidation, timeout, and concurrency tests.
Payment mismatch or duplication	High	Verified callbacks, idempotency, immutable transaction history, reconciliation, and alerts.
Account compromise	High	Strong authentication, rate limiting, session control, monitoring, optional MFA, and incident response.
Data loss	Critical	Automated backups, protected storage, restore testing, retention, and recovery plan.
Subscription abuse	Medium	Server-side feature enforcement, limits, billing validation, and audit.
Incorrect venue data	Medium	Vendor responsibility, required fields, publication validation, moderation, and customer reporting.
External service outage	High	Timeouts, retries, queues, fallback messages, circuit breaking where appropriate, and status monitoring.
Scope expansion	High	Phase-1 boundaries, MoSCoW priorities, change control, and separate future roadmap.

Risk	Impact	Mitigation
Low vendor adoption	High	Affordable configurable plans, simple onboarding, training, support, and measurable pilot rollout.

19. Future Roadmap

Stage	Potential capabilities
Post-launch stabilization	Performance optimization, stronger reports, operational automation, support improvements, additional local payment options.
Growth release	Additional sports, cities, multilingual interface, loyalty, referral, recurring booking, group booking, richer CRM and marketing.
Ecosystem release	Tournament management, teams, player communities, coaching services, equipment rental, academy services, and event management.
Intelligence release	AI-assisted recommendations, intelligent scheduling, demand insights, fraud detection, and advanced analytics.
Channel expansion	Native mobile applications, partner APIs, corporate packages, and broader geographic deployment.

20. Requirement Traceability and Glossary

20.1 Traceability Convention

Prefix	Meaning
G	Product goal
FR-IAM	Identity and access requirement
FR-MKT	Marketplace requirement
FR-TEN / VEN / GRD	Tenant, venue, and ground requirement
FR-PRI / AVL	Pricing and availability requirement
FR-BKG	Booking requirement
FR-PAY	Payment and refund requirement
FR-FIELD / STAFF	Field and staff operations requirement
FR-SUB	Subscription requirement
FR-ADM	Super Admin requirement
FR-RPT / NTF	Reporting and notification requirement
BR	Business rule
NFR	Non-functional requirement
USR / TAD / FAD / SAD	User, Tenant Admin, Field Admin, and Super Admin scenario
AC	Release acceptance criterion

20.2 Compact Traceability Matrix

Capability	Goals	Requirements	Scenarios	Acceptance
Discover and book venues	G-001, G-002, G-005	FR-MKT, FR-AVL, FR-BKG, FR-PAY	USR-01 to USR-08	AC-001 to AC-003
Operate tenant business	G-003, G-004, G-006	FR-TEN, FR-VEN, FR-GRD, FR-PRI, FR-STAFF	TAD-01 to TAD-10	AC-004, AC-005
Run field operations	G-003, G-005	FR-FIELD, FR-AVL, FR-BKG	FAD-01 to FAD-07	AC-006, AC-007
Govern platform	G-004, G-007, G-008	FR-ADM, FR-SUB, FR-PAY, FR-RPT	SAD-01 to SAD-07	AC-008 to AC-012
Protect platform	G-004, G-008	FR-IAM, NFR-SEC, audit rules	All protected scenarios	AC-002, AC-005, AC-006, AC-011, AC-012

20.3 Glossary

Term	Definition
Tenant	An independent sports business using the MeroMaidan SaaS workspace.
Tenant Admin / Company Admin	The owner or authorized administrator with tenant-wide control.
Field Admin	A tenant-scoped operational administrator for assigned venues or grounds.
Ground	A bookable field, court, pitch, or playing area within a venue.
Marketplace	The public venue-discovery and booking area.
RBAC	Role-Based Access Control; permissions are assigned through roles.
Tenant context	The active tenant identity used to scope data and authorization.
Booking lock	Temporary protection of a selected slot while validation or payment is completed.
Idempotency	Processing repeated requests safely without creating duplicate effects.
CRM	Customer Relationship Management.
ERP	Enterprise Resource Planning for operational resources such as equipment and maintenance.
CMS	Content Management System for news, events, gallery, announcements, and pages.
MRR	Monthly Recurring Revenue.
Audit log	Protected historical record of important user, business, security, and administrative actions.

Final document status: This consolidated PRD is ready for structured stakeholder review. Items explicitly identified as decisions required must be finalized before development baselines and contractual acceptance are approved.